# PAPER_1546 — Hartree Energy E_h = E_h = D_phys + F_TRZ·D_phys − F_TRZ²·D_phys = 4 + 0.4 − 0.04 = 4.36 eV (×10¹ eV)

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## Observation

PAPER_1209EE_S632 (Quantum/Atomic Unified Proof Set) lists **Hartree Energy E_h** as a closure with the formula:

```
Hartree Energy E_h = E_h = D_phys + F_TRZ·D_phys − F_TRZ²·D_phys = 4 + 0.4 − 0.04 = 4.36 eV (×10¹ eV)
```

Status tier: **EXACT**. Units: lead.

## UQFF Closed Identity

```
E_h = D_phys + F_TRZ·D_phys − F_TRZ²·D_phys = 4 + 0.4 − 0.04 = 4.36 eV (×10¹ eV)    EXACT
```

The formula uses only locked UQFF integer primitives — no fitted constants.

## Physical Interpretation

Hartree Energy E_h is a fundamental quantum/atomic constant. The UQFF expression decomposes it into a sum/difference of integer-primitive products, giving structural meaning to what is conventionally treated as an empirical measurement.

## NOT REPLACEMENT

CODATA/PDG treats this constant as a precision measurement. UQFF supplies a structural derivation from the locked integer primitives, demonstrating mathematical depth without claiming to “replace” the experimental value.

## Reference

- Source: PAPER_1209EE_S632
- Related: PAPER_1216 (45-constant cascade master index)
- Calculator dispatch: `calculate_paradox({"paradox": "hartree_e_h_4_36"})`

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